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Education

Singapore University of Technology and Design, Bachelor of Engineering in Computer Science, 2026.

- **Cocurriculars**: Galois Group Mathematics Society, Google Developer Student Club (3DC).
- **Concentrations**: Minor in Artificial Intelligence, Specialisation in Visual Analytics and Computing.

Experiences

Creatz3D and Figure Factories, AI Engineer, 2025.

1. Accelerated mesh fixing by designing, training, and testing novel GNNs to thicken meshes under geometric constraint.
2. Operationalised mesh creation by deploying cron-scheduled prompt-to-image-to-mesh generation pipelines.
3. Streamlined UGC production by orchestrating human-in-the-loop generative AI workflows.

Proficiencies

- **Languages.** C, CSS, Dockerfile, HTML, Java, JavaScript, Makefile, Python, Rust, Shell, SnekScript, Swift, (La)TeX.
- **Tools.** AWS, Azure, Cloudflare, Docker, FAISS, FastAPI, Figma, Firebase, Flask, Git(Hub), Gradio, Hugging Face, LangChain, llama.cpp, Matplotlib, MLX, Next.js, NumPy, Ollama, OpenRouter, Open WebUI, Paint.NET, PyTorch, Qdrant, React, SQL, Shadcn, Streamlit, Tailwind, TensorFlow, Vercel, Whisper, n8n, pandas, scikit-learn.

Projects

PDPal (capstoneshowcase.sutd.edu.sg/project/proj-07-asiacloud-ai4pdpa), AI Legal Assistant, 2026.

1. Crafted backend architecture by defining targets, running experiments, and reasoning over trade-offs with team.
2. Tested and observed system outperform GPT-5.4 by 4.5% and Gemini 3.1 Pro by 9.1% on YPDYC oracle@3.
3. Improved frontend accessibility by designing responsive multilingual visual components.
4. Validated frontend accessibility with 73% of users fully recommending product in preliminary studies.

SUTDoko (sutdoko.vercel.app), SUTD Room Address Lookup Tool, 2026.

1. Trained Chinchilla-optimal Nanochat-style transformers to denoise SUTD room names.
2. Tested and observed model outperform SUTD Assistive Maps on test subset accuracy.
3. Optimised inference smoothness and speed by making it cached, debounced, and incremental.

Snekboros, "Vision World Models for Snake Playing", 2026.

1. Trained SIGReg-regularised convolutional GRU world models to play 1P Snake purely from 4x4 grid images.
2. Tested and observed model underperform oracle by 11% and outperform random baseline by 89% on test set score.
3. Observed effect of planning horizon and sample count on test set score.

Vibe Decoding, "An Unexpected Expected Next-Token Rule for Language Models", 2025.

1. Proposed aligning next-token embeddings with their expectation as next-token rule for language models.
2. Tested and observed Qwen, Gemma, and Llama models improve on BoolQ and NumerSense macro F1 by up to 7.7%.
3. Characterised asymptotic runtime and greedy interval of rule.

BLIPS Regularisation, "Regularisation with Buffered Linear Interpolation by Positional Similarity", 2025.

1. Proposed using buffer to linearly interpolate between recent feature maps by positional similarity.
2. Trained and tested CNNs, RNNs, and CNN-ViT hybrids on train subset.
3. Tested and observed best model achieve 99.6% macro F1 on test subset.
4. Interpreted failures of best model by visualising Grad-CAM attention on incorrect classifications.

Recognition

- Showcased MLOps Projects, Singapore University of Technology and Design, 2026.
- Showcased Deep Learning Projects, Singapore University of Technology and Design, 2025.
- Information Systems Prize, Singapore University of Technology and Design & Singtel, 2024.
- Mathematics and Science Book Prize, Temasek Secondary School, 2017.